

CHAPTER 4

MATERIALS, METHODS AND CHARACTERIZATION TECHNIQUES

This chapter provided a brief overview of the ingredients and process utilized to create magnesium oxide (MnO) nanoparticles (NPs) using more environmentally friendly methods. Additionally, this part provided a brief explanation of the fundamental characterization approaches used in the current work.

4.1 MATERIALS AND METHODS

4.1.1. Materials for synthesis of Pure MnO nanoparticles via biological routes

Manganese sulfate (MnSO_4) Salts, sodium hydroxide (NaOH) and Congo Red dye are purchased from mercury scientific chemical Limited, India. De-ionized water was used throughout this study. For biological or green synthesis, 5 kg of *Alstonia scholaris* flower was harvested from agriculture field in nearby area in Erode, Tamilnadu.

4.1.2. Green synthesis procedure of Pure MnO(MSC-NPs) nanoparticles

A 500 g of *Alstonia scholaris* flower was collected and cleaned, chopped small, and then prepared in deionized water. For future research, the resultant extract was kept in an airtight container. A standard process was used for and Syzygium Cumini leaves extract. The 1M of MnSO_4 salt solution was stirred continuously for 20 minutes. For three hours, the mixture was constantly agitated at 800 rpm. Then NaOH solution is added drop by drop with constant stirring. When the reaction was finished, a brown colour precipitate was produced. After centrifuging the resultant precipitate, de-ionized water was used for washing. Using ethanol and water, the washing and centrifuging process was carried out multiple times. The collected samples were dried in a hot air atmosphere for 8 to 9 hours at 70 degrees Celsius. The precipitate was named as MAS-NPS is showed in Figure.1

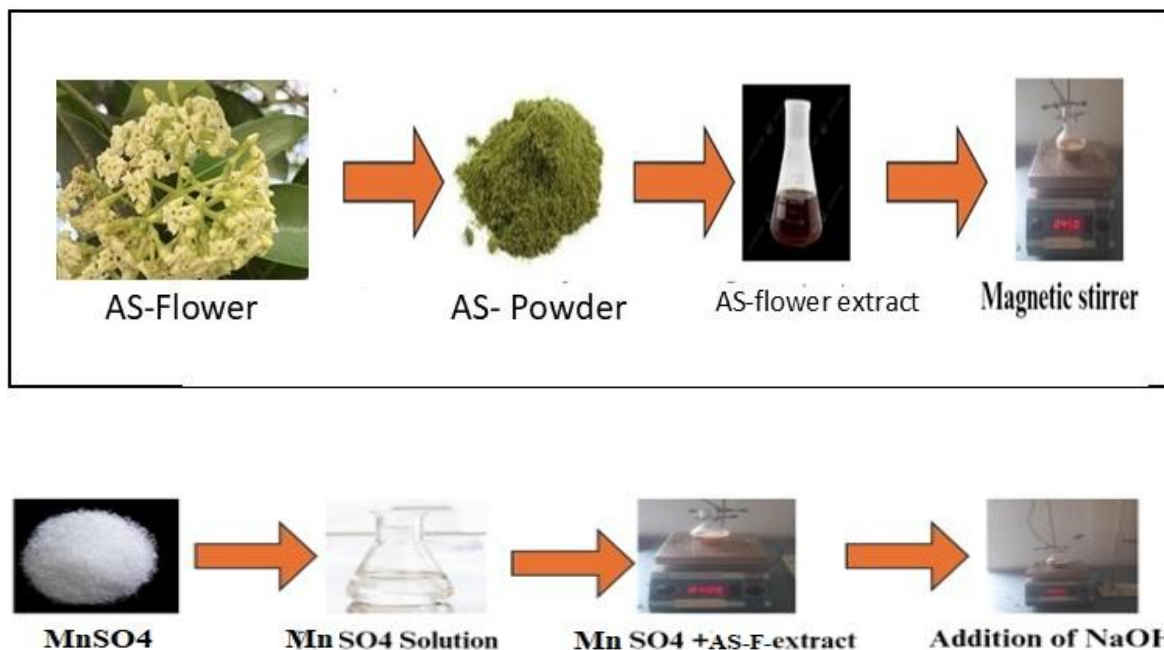


Figure 1. Schematic representation synthesis of MAS-NPs from biological route

4.2. CHARACTERIZATION TECHNIQUES

4.2.1 X-Ray Diffractometer (XRD)

The prepared material's structural properties could be investigated using the well-known X-ray diffraction method. This technique has been used to detect stress and strain, chemical phase analysis, phase equilibrium, particle size measurement and crystal structure identification. The synthesised samples were characterized by X-ray diffraction (Bruker) with CuK α radiation ($\lambda = 1.5418 \text{ \AA}$). A diffraction pattern has been observed when a highly collimated beam of X-rays falls on material due to diffraction from the crystalline phases of the sample. The obtained diffraction pattern is used to determine the materials and their structural properties. Using 1st order Bragg's equation, one can easily find the lattice spacing from this diffraction pattern.

$$n\lambda = 2d \sin\theta \quad \text{-----} \quad (1.1)$$

Where λ is the electron wavelength, d is the interplanar distance, and θ is the glancing angle. The intensity of diffracted X-rays is measured in terms of the angle of diffraction 2θ and the sample orientation. The Scherer formula, which is derived from the diffraction peaks, is used to calculate the crystalline size.

$$D = \frac{0.9 \lambda}{\beta \cos \theta} \quad \text{----- (1.2)}$$

Here, λ is the wavelength of the X-ray wavelength source (in nm) and β is Full-Width Half Maximum (FWHM) (in degree) [39] .

The X-ray Diffractometer (XRD) is shown schematically in Figure.2

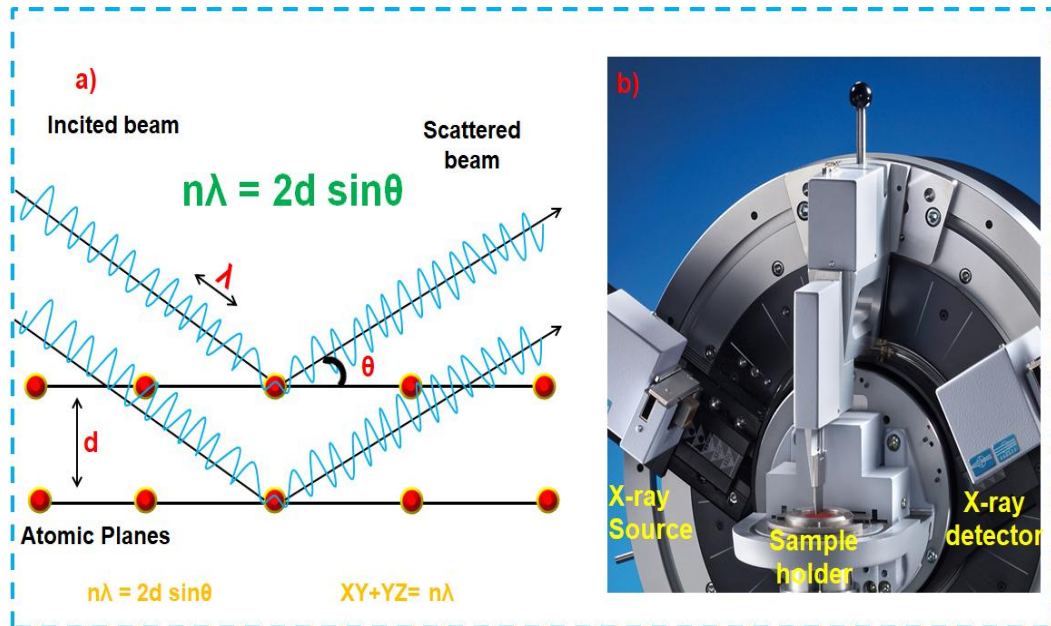


Figure.2 Schematic representation of Bragg's X-ray diffraction and (b) X-ray diffractometer

4.2.2 Field Emission Scanning Electron Microscope (FESEM)

FESEM images were taken with a field emission scanning electron microscope (JEOL Model JSM - 6390LV) and energy dispersive spectra (EDS) were taken with an energy dispersive spectra (JEOL Model JSM - 6390LV) (JEOL Model JED e 2300). FESEM images were used to determine the surface morphology of the samples. The instrument's resolution was 3 nm. The accelerating voltage is scalable in 55 steps from 1.5 to 30 kV. The magnification could be adjusted from 12 - 106 X. The current range of a probe was 4 pA to 10 nA. Figure 3 shows a schematic diagram of a scanning electron microscope [40].

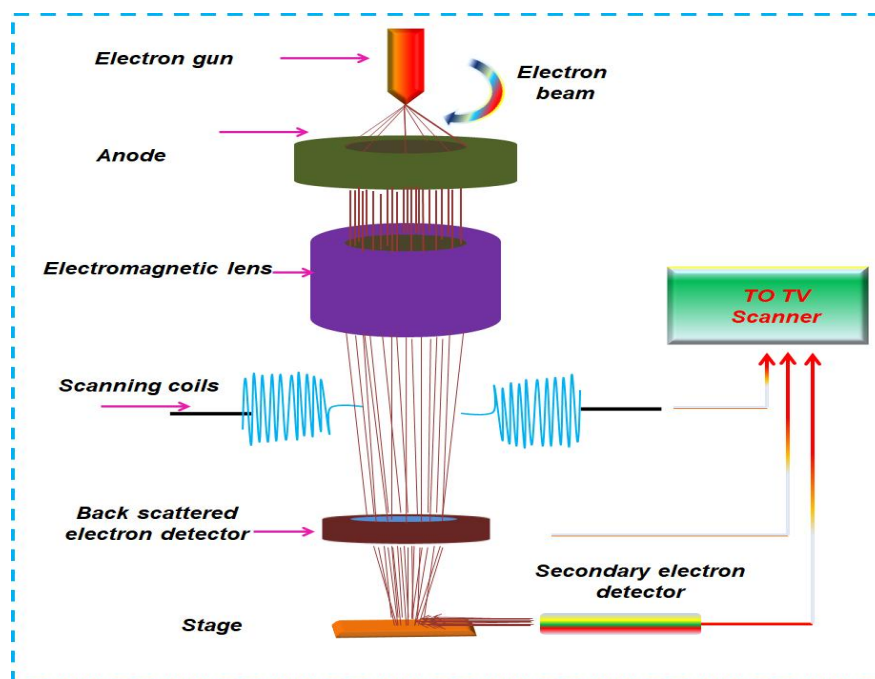


Figure.3 Schematic diagram of Scanning Electron Microscope

4.2.3. Energy Dispersive X-Ray Analysis (EDX)

EDX instrument is an important tool to identify the presence of elemental (or) chemical composition of prepared sample. Figure 4 shows an experimental set up of EDX spectrum which consists of four major components such as

1. The excitation source consisting of an electron beam or x-ray beam
2. The X-ray detector
3. The pulse processor
4. The analyzer

EDX systems are a combined system integrated with either EPMA or SEM instrument. The electron beam energy is generated between the requirements of resolution and X-ray production efficiency. The two fully focusing crystal spectrometers are used to focus X-radiation excited in the specimen. The EDX-ray detector was used to find out the relative abundance of emitted X-rays versus their energy. EDX systems consist of a liquid nitrogen Dewar for cooling, sensitive X-ray detector and software to collect and interpret energy spectra. At the end of a long arm of main instrument the detector is mounted on a sample chamber which is cooled by themselves by liquid nitrogen. Si(Li) crystals is the most common detectors which can operate at low voltages but the advanced "silicon drift detectors" can operate at higher count rates without cooling using liquid nitrogen. Crystal present in EDX detector yields free electrons by absorbing the energy of the incoming x-rays by ionization. The absorbed electrons become conductive and produce electric charge. The detector has typically lithium drifted silicon, solid state device. When an incident X-ray strikes the detector, it creates a charge pulse that is proportional to energy of X-ray. By a charge-sensitive preamplifier the charge pulse is converted to a voltage pulse. After that the signal is sent to a multichannel analyser where the pulses are sorted by voltage. The x-ray absorption by crystals converts energy of each individual rays into electrical voltages of proportional size, the electrical pulses correspond to the characteristic x-rays of the element. From the voltage measurement, the energy was estimated for each incident X-ray which was sent to a computer for display and further data evaluation. The elemental composition of the sample volume was determined by the spectrum of X-ray energy versus counts [41].

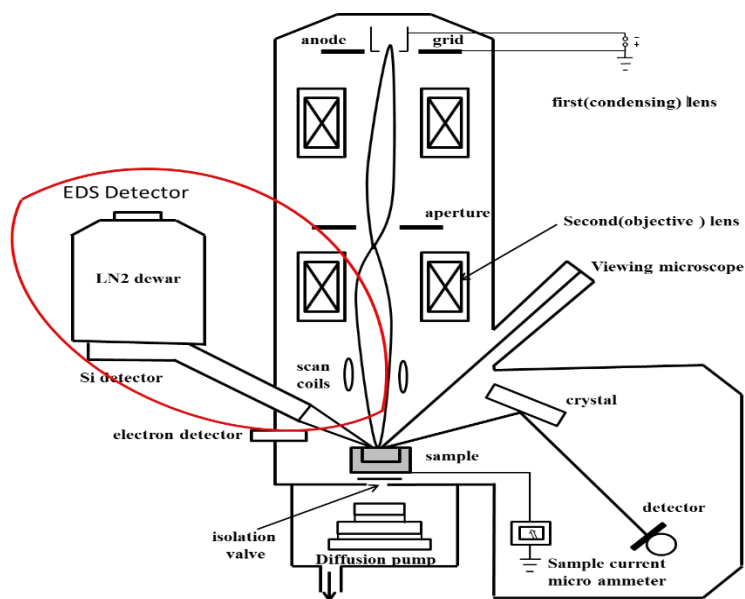


Figure 4. Schematic diagram of Energy Dispersive X-ray Spectroscopy (EDS)

4.2.4 Fourier-Transform Infrared spectroscopy (FT-IR)

FT-IR spectrometer is used to determine the presence of vibrations in the synthesized materials.

Instrumentation

The FT-IR spectrometer instrument consists of an IR source, beam splitter, interferometer, detector, sample compartment, amplifier, A/D converter and a computer. The source generates radiation which passes the sample through the interferometer and reaches the detector. After that the signal is amplified and converted to a digital signal by the amplifier and analog-to-digital converter, respectively. Eventually, the signal is transferred to a computer in which Fourier transform is carried out. The radiation generated by the source passes into the sample through the interferometer which is a combination of mirrors, source and beam splitter and electric signal reaches the detector. Using amplifier and analog-to-digital converter the output signal is amplified and converted to digital signal. Ultimately, the digital signal is transmitted to a computer in which Fourier transform function is carried out. An interferometer is used in a FT-IR spectrometer for the modulation of wavelength from

a broad band source. The intensity of the transmitted or can be considered as a rejected light is measured by a detector as a

function of its wavelength. The signal received from the detector is an interferogram a background collection of spectrum along with a single-beam spectrum of the sample and analysed with a computer and transferred into frequency data by using Fourier transforms process. Finally the FT-IR spectra are plotted as intensity versus wavenumber (in cm^{-1}). Wavenumber is the reciprocal of the wavelength. The intensity can be presented as the percentage of light transmitted or absorbed at each wavenumber.

Using pressed pellet technique, solid sample with a lesser amount of 0.1 to 1.0 % is mixed with 200 to 250 mg of potassium bromide, ground finely to form a homogenous mixture. The mixture is compressed into a thin transparent pellet using a hydraulic press using a force of approximately 8 tons under a vacuum for several minutes. These prepared pellets should be transparent to IR radiation spectral analysis [42].

Now lets look an FT instrument

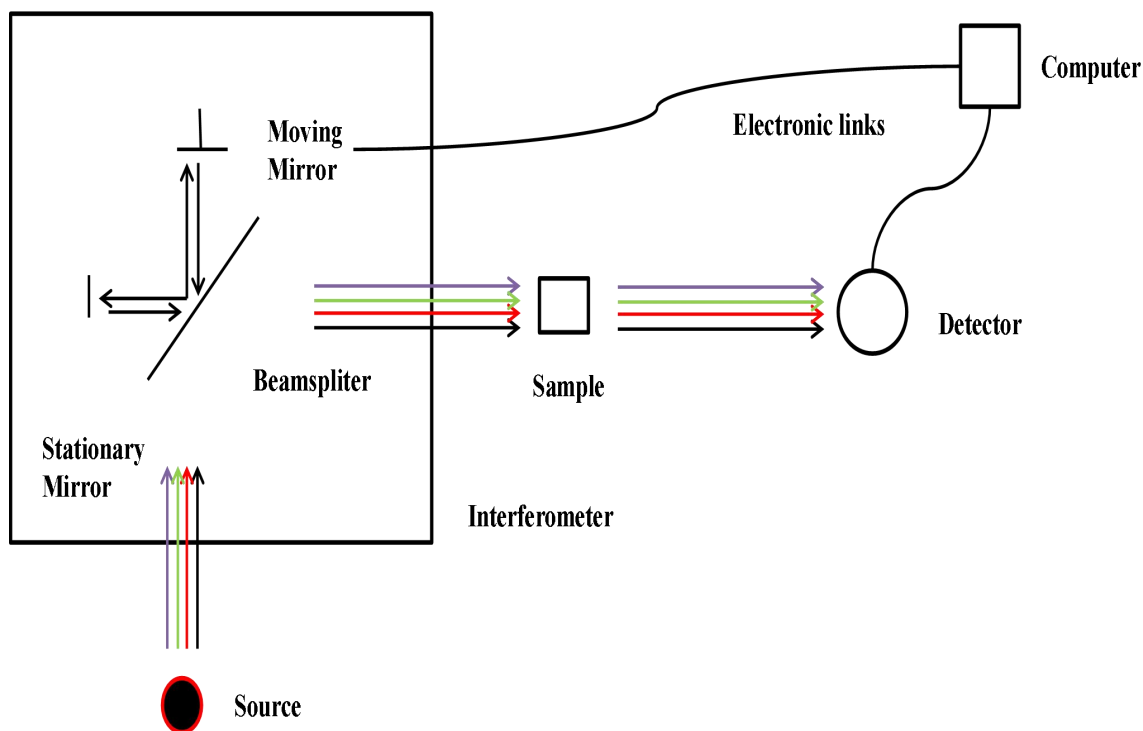


Figure 5. Schematic representation of FT-IR

4.2.5 Ultraviolet–Visible Spectroscopy

Principle

The solid, liquid and gases samples are analysing through spectrophotometer by the use of radiant energy which having the wavelengths for Ultra–Violet (UV), Visible (Vis) and near infrared (NIR) radiation ranges are 300 to 400 nm, 400 to 765 nm and 765 to 3200 nm, respectively.[50]. It is an absorption spectroscopy in which various electric transitions are occurred. More number of electrons are excited from HOMO (highest occupied molecular orbital) to LUMO (lowest unoccupied molecular orbital) when the molecules are irradiated with light energy. The wavelength and intensity are recorded by optical spectrometer when the absorption happens [43].

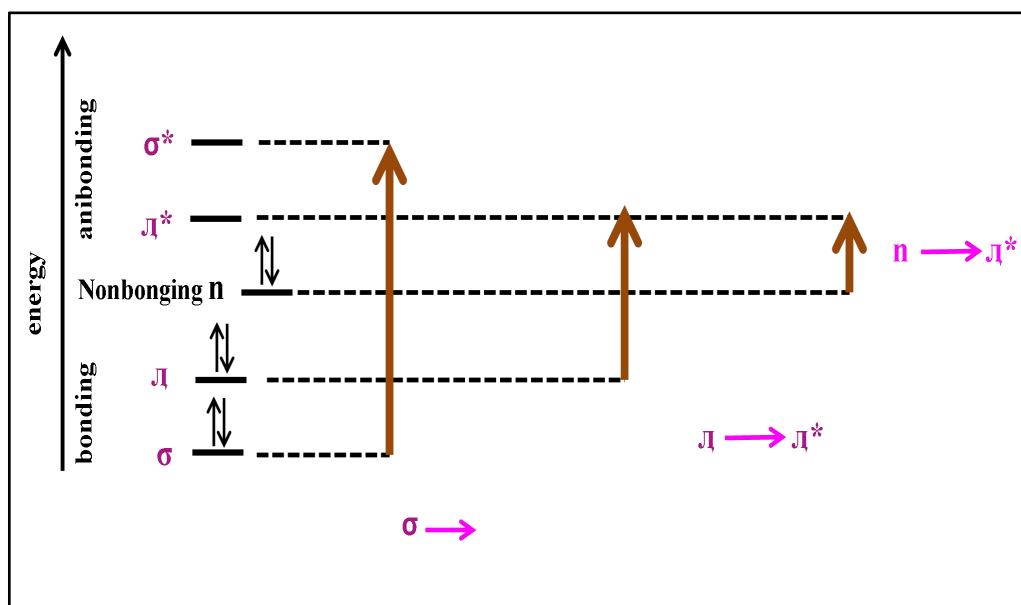


Figure 6. Energy level diagram of a molecule

UV spectroscopy obeys the Beer-Lambert law. It states that, when a beam of monochromatic radiation passes through a solution of absorbing substance the rate of decrease of intensity of radiation with thickness of absorbing solution is proportional to the intensity of incident radiation and concentration of the solution.

The expression of Beer-Lambert law is

$$\log (I_0/I) = E Cx \quad \text{-----}(1.3)$$

Where, $\log (I_0/I) = A$ = absorptivity

I_0 = intensity of incident light

I = intensity of light leaving from sample cell

C = molar concentration

x = path length (cm.)

$E = k/2.303$ = molar absorptivity or molar extinction co-efficient

k = molar absorption co-efficient.

The absorption indices or extinction coefficient (k) can be calculated making use of Lambert's equation

$$I = I_0 \exp \frac{(-4\pi kd)}{\lambda} \quad \text{-----}(1.4)$$

where, I/I_0 is the transmittance through which the electromagnetic wave travels to change its intensity from I_0 to I , λ is the wavelength of the light and d is the thickness of the material. The direct energy band gap (E_g) of the synthesized materials was calculated by using the Tauc's plot,

$$(\alpha h\nu)^n = B^*(h\nu - E_g) \quad \text{-----} (1.5)$$

Where, α & $h\nu$ represents the absorbance coefficient ($\alpha = 2.303A/l$) and photon energy, E_g – is the optical band gap, B - band tailing parameter and n - representing the optical transition, which have some specific values (i.e) $1/2$, 2 , $3/2$ (or) 3 respectively for direct, indirect, forbidden direct (or) forbidden indirect band gap. The direct band gap was plotted from the given relation

$$(\alpha h\nu)^{1/2} \text{ Vs } (h\nu) \quad \text{-----}(1.6)$$

A beam of light from UV or visible source is partitioned into specific component wavelength diffraction grating or prism. Monochromator splits each single beam of wavelength into two beams of equal intensity by a half-mirrored device. One of the sample beam with magenta colour passes through cuvette, a small transparent container containing a solution of the analysing sample in a transparent solvent. The other blue coloured beam is a reference beam passing through the solvent containing cuvette. The intensities of both of these lights are measured and compared by electronic detectors. I_0 is the intensity of the reference beam without any suffering or light absorption whereas I is the intensity of the sample material. Within a short duration the spectrometer starts scanning automatically all the constituent's wavelength in a specified manner. The scanning region of UV portion is normally from 200 to 400 nm and the visible portion is from 400 to 800 nm [44].

Instrumentation

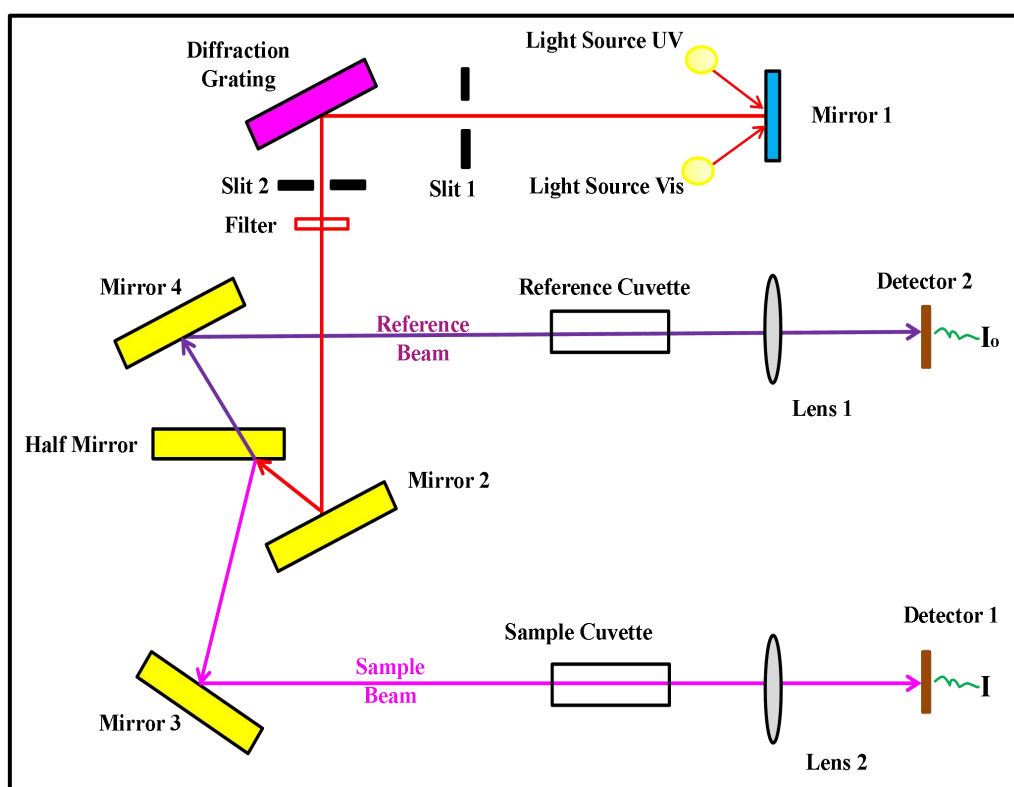


Figure 7. Schematic representation of UV

A beam of light from a visible and/or UV light source (coloured red) is separated into its component wavelengths by a prism or diffraction grating. Each monochromatic (single wavelength) beam in turn is split into two equal intensity beams by a half-mirrored device. One beam, the sample beam (coloured magenta), passes through a small transparent container (cuvette) containing a solution of the compound being studied in a transparent solvent. The other beam, the reference (coloured blue), passes through an identical cuvette containing only the solvent. The intensities of these light beams are then measured by electronic detectors and compared. The intensity of the reference beam, which should have suffered little or no light absorption, is defined as I_0 . The intensity of the sample beam is defined as I . Over a short period of time, the spectrometer automatically scans all the component wavelengths in the manner described. The ultraviolet (UV) region scanned is normally from 200 to 400 nm, and the visible portion is from 400 to 800 nm.

When the sample absorb light the wavelength is given as $I < I_0$ however $I = I_0$ is given when the sample does not absorb light. The difference between I and I_0 is plotted as graph versus wavelength. Absorption may be represented as absorbance ($A = \log I_0/I$) or transmittance ($T = I/I_0$). The wavelength of maximum absorbance is a characteristic value, designated as λ_{max} . Different compounds exhibit different absorbances and absorption maxima. Compounds absorbing intensely must be studied in dilute solution, so the detector can receive significant light energy and this requires the use of non-absorbing completely transparent solvents. The most commonly used solvents are water, cyclohexane, ethanol and hexane. Solvents containing double or triple bonds, or heavy atoms (e.g. S, Br & I) are commonly avoided.

Hydrogen-Deuterium lamps and tungsten filament lamps with emission radiations of 375 nm and Hydrogen-Deuterium lamps with emission radiation below 375 nm are most widely used and suitable light source. Monochromators generally composed of prisms and slits. The most of the spectrophotometers are double beam spectrophotometers. Both sample and reference solution are contained in the cells made up of quartz or silica. Two photocells receiving beam from sample cell and detector receives beam from reference cell. Amplifier is coupled to a pen recorder which is coupled to the computer [45].

4.2.6. Photocatalytic Experiment

At room temperature, the photodegradation of the prepared material was tested against aqueous solution of Congo Red(CR) dye under UV light irradiation. Where, 0.2 g of prepared Nano catalysts was added into the 30 ml of 10 μ M dye in aqueous solution. To ensure the adsorption-desorption equilibrium in the catalyst and dye, the mixture of solution was made to stir under dark condition for an hour. Then the solution was kept under UV light irradiation and subjected to degradation analysis with respect to different time intervals (30 mins, 60 mins, 90 mins, 120 mins). The basic experimental set up of the photocatalytic experiment was shown in figure 2.16.

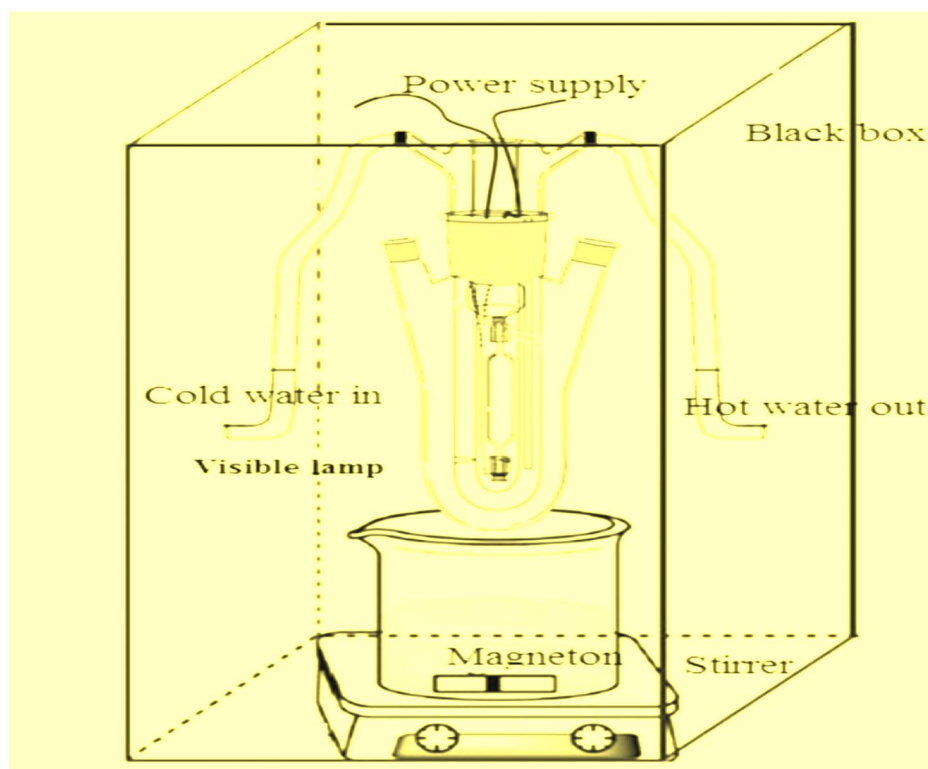


Figure.8. Schematic diagram of experimental setup used in the photocatalytic degradation

From the concentration plot, the dye degradation efficiency was calculated using the following formula

$$\% \text{ Degradation} = \left(\frac{C_0 - C_t}{C_t} \right) \times 100 \quad (1.7)$$

Where, C_0 - Absorption intensity of initial dye solution

C_t - main absorption peak intensity dye at certain time [46].

4.2.7 ANTI MICROBIAL ACTIVITIES

Preparation of the Bacterial Inoculum

Stock cultures were maintained at 4° C on slopes of nutrient agar and potato dextrose agar. Active culture for experiments were prepared by transferring a loop full of cells from stock cultures to test tubes of 50ml nutrient broth bacterial cultures were incubated with agitation for 24 hours at 37°c on shaking incubator and fungal cultures were incubated at 27°c for 3-5 days. Each suspension of test organism was subsequently stroke out on nutrient agar media and potato dextrose agar. Bacterial cultures then incubated at 37°c for 24 hours and fungal incubated at 27°c for 3-5 days. A single colony was transferred to nutrient agar media slants were incubated at 37°c for 24 hours and potato dextrose slant were incubated at 27°c for 3-5 days. These stock cultures were kept at 4°c. For use in experiments, a loop of each test organism was transferred into 50ml nutrient broth and incubated separately at 37°c for 18-20 hours for bacterial culture.

Well Diffusion method

The antibacterial activity and antifungal activity of crude extract extracts was determined by Well Diffusion method.[54] (Bauer *et al.*, 1996).. The 2-20 µl of Nanoparticle extract was poured into the wells. After that, the plates were incubated at 37°C for 24 hours. Assay was carried out in triplicates and control plates were also maintained. Zone of inhibition was measured from the edge of the well to the zone in mm. The tested cell

suspension was spread on Mueller Hinton Agar plate and Potato Dextrose Agar plates. Wells were put into the agar medium using sterile forceps. Plant extract was poured on to wells. Then plates were incubated at 37°C for about 24 hours and control was also maintained. Zone of inhibition was measured from the clear zone in mm.

Antibacterial activity was performed by using Agar Diffusion Method (Van der Watt *et al.*, 2001).[55] The stock cultures of bacteria (*E. coli* and *Streptococcus*) were received by inoculating in nutrient broth media and grown at 37 °C for 18 hours. The agar plates of the above media were prepared. Each plate was inoculated with 18 hours old cultures the bacteria were swab in the sterile plates. Cut the 5 wells Pour the extract in ratio 25 µl, 50 µl 75 µl 100 µl. All the plates were incubated at 37°C for 24 hours and the diameter of inhibition zone was noted in Cm.

Agar well diffusion method has been used to determine the antimicrobial activities and minimum inhibitory concentrations of plant extracts against Gram-positive, Gram-negative bacteria. The extracts exhibited antibacterial activities against tested microorganisms [47,48].

4.2.8 ANTIOXIDANT ACTIVITY

DPPH radical scavenging activity

The free radical scavenging activity of methanolic extract of plant extract was measured by using 2, 2-diphenyl-1-picrylhydrazyl (DPPH) The scavenging activity for DPPH free radicals was measured according to the procedure described by (Braca *et al.*, 2001). An aliquot of 3 ml of 0.004% DPPH solution in methanol and 0.5 to 2.5 µl of plant extract/ascorbic acid at various concentrations were mixed. The mixture was shaken vigorously and allowed to reach a steady state at room temperature for 30 min. Decolorization of DPPH was determined by measuring the absorbance at 517 nm. A control was prepared using 0.1 ml of respective vehicle in the place of plant extract/ascorbic acid. The percentage inhibition of DPPH radicals by the extract/compound was determined by comparing the absorbance values of the control and the experimental tubes [49].

$$\text{Scavenging activity \%} = \frac{\text{A518 (control)} - \text{A518 (sample)}=045}{\text{A518 (control)}} \times 100$$